

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M. Sc. (Biotechnology/Microbiology/Biochemistry) Examination, November 2018

Course code & Name: MS 811 & Omics and Bioinformatics

Date: 19/11/2018 Day: Monday Time: 1:30 P.M. To 2:00 P.M. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	20
1.	The following is a fourth-generation “post light” sequencing method that utilizes semiconductors to measure the release of protons (H^+) when a new dNTP is added to a growing strand of DNA.	01
	(i) Oxford nanopore	(ii) Illumina/Solexa method
	(iii) 454 Pyrosequencing	(iv) Ion torrent
2.	In order to construct genomic library of a bacterial genome (4×10^6 bp) which has been fragmented to obtain an average insert size of 0.2 Kb. What will be the appropriate number of clones that you will require to obtain three-fold sequence coverage of the genome?	01
	(i) 12000 clones	(ii) 16000 clones
	(iii) 60000 clones	(iv) 20000 clones
3.	Long interspersed nuclear elements (LINEs) and short interspersed nuclear elements (SINEs) are the two categories of _____.	01
	(i) Tandem repeats	(ii) Non-LTR Retrotransposons
	(iii) LTR-DNA transposons	(iv) Pseudogenes

4.	The following sequencing by ligation method utilizes a set of 16 different oligonucleotides (octamers) tagged at the 5' end with a different fluorescent dye to identify the dinucleotide positions on the template sequence.		01
	(i) HeliScope sequencing	(ii) PacBio SMRT Sequencing	
	(iii) SOLiD sequencing	(iv) 454 Pyrosequencing	
5.	In radiation-hybrid genome mapping, the recipient cell lines (usually hamster or mouse) are deficient either for _____ or _____ genes.		01
	(i) HAT and CYP2D6	(ii) LacZ and Amp ^R	
	(iii) HIS3 and LEU2	(iv) TK and HGPRT	
6.	The following evolutionary model used to infer molecular phylogeny assumes that transitions occur more frequently than transversions?		01
	(i) Jukes-Cantor	(ii) Minimum Evolution	
	(iii) Maximum Likelihood	(iv) Kimura	
7.	The genetic polymorphism in the human _____ gene encoding an ATP-dependent membrane efflux pump (P-glycoprotein) affects pharmacokinetic parameters by influencing the plasma concentrations of certain drugs.		01
	(i) MDR1	(ii) RAR α	
	(iii) ATRA	(iv) CYP450	
8.	Which of the following quantitative proteomics method utilizes proteolytic digestion of two distinct proteomes in aqueous buffer containing either normal water (H ₂ ¹⁶ O) or isotopically labelled water (H ₂ ¹⁸ O).		01
	(i) Metabolic stable-isotope coding	(ii) Isotope-coded affinity tag	
	(iii) Enzymatic stable-isotope coding	(iv) Protein microarrays	
9.	Which of the following is a workflow software offering set of various tools for the analysis of RNA-Seq data?		01
	(i) Uniprot	(ii) STAMP	
	(iii) VAST	(iv) Tuxedo	
10.	The following technique utilizes biphasic microcapillary column packed with strong cation exchange (SCX) and reverse-phase (RP) materials for LC-MS/MS analyses of tryptic digests of proteome samples in order to separate and identify individual proteins.		01
	(i) 2D-DIGE	(ii) Protein microarrays	
	(iii) MudPIT	(iv) Yeast-two hybrid	
11.	The following comparative genomics method is used to predict functionally linked proteins that may be involved in protein-protein interactions.		01
	(i) Epistasis	(ii) Rosetta stone	
	(iii) FRET	(iv) Phage display	
12.	Photolithography method utilizes optical systems and _____ chemistry for selective extension of nucleotide polymers to manufacture DNA microarrays.		01
	(i) Reversible terminator	(ii) Cy3 N-hydroxysuccinimidyl ester	
	(iii) 2', 3'-dideoxynucleotide	(iv) Phosphoramidite	
13.	The following technique utilizes isoelectric focusing followed by (SDS)-polyacrylamide gel electrophoresis at right angles to separate differentially expressed proteins within proteome samples.		01
	(i) 2D-LC/MS	(ii) 2D-PAGE	
	(iii) 2D-DIGE	(iv) Both (ii) & (iii)	

14.	In metabolome analysis, which of the following analytical technique is used for separation and identification of metabolites?		01
	(i) GC-MS	(ii) CE-MS	
	(iii) LC-MS	(iv) All of them	
15.	In a yeast two-hybrid system, the transcription of _____ reporter gene is activated upon interaction between the bait and prey proteins.		01
	(i) Gal4	(ii) LexA	
	(iii) HIS3	(iv) All of them	
16.	The following is a statistical resampling method used to evaluate the reliability of an inferred phylogenetic tree.		01
	(i) Bootstrapping	(ii) Neighbour joining	
	(iii) UPGMA	(iv) Maximum parsimony	
17.	High-throughput whole genome pyrosequencing utilizes _____ to enable genomic DNA fragments to be attached to small metallic beads coated with streptavidin.		01
	(i) Bridge amplification	(ii) 5'-biotin labeled adapters	
	(iii) Iodoacetamide reagent	(iv) 3'-Cy5-labeled adapters	
18.	The following carbohydrate-binding protein with affinity for different glycans is used in sandwich-style assay to detect glycosylated proteins on protein microarrays.		01
	(i) Anti-phosphotyrosine antibody	(ii) Biotinylated lectin	
	(iii) Cy5-labelled peptidoglycan	(iv) Biotinylated trypsin	
19.	The following scheme is used for classification of protein structures.		01
	(i) SCOP	(ii) CATH	
	(iii) PROCHECK	(iv) Both (i) and (ii)	
20.	Which of the following is a molecular graphics software used for the visualization of protein structures?		01
	(i) MolScript	(ii) RasMol	
	(iii) Ribbons	(iv) All of them	

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III Semester of M. Sc. (Biotechnology/Microbiology/Biochemistry) Examination, November 2018

Course code & Name: MS 811 & Omics and Bioinformatics

Date: 19/11/2018 Day: Monday Time: 2:00 P.M. To 4:30 P.M. Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q – II Answer the following questions as directed (answer any five).		20
1.	Explain the synthesis of DNA microarrays on a glass substrate using photolithography.	04
2.	Explain high throughput genome sequencing using reversible terminator technology.	04
3.	Explain the hierarchical shotgun sequencing strategy used in the human genome project.	04
4.	Explain the pyrosequencing reactions and massively parallel pyrosequencing approach.	04
5.	Explain the radiation hybrid mapping of the human genome using STS markers.	04
6.	Define metagenomics. Explain briefly the sequence-driven and expression-driven metagenomic approaches.	04
7.	Give an account of information-flow in pharmacogenomics. Explain how genetic polymorphisms affect clinical outcomes of drugs in different individuals.	04
SECTION - II		
Q-III Answer the following questions as directed (answer any ten).		30
1.	Describe briefly the main analytical techniques used in metabolome analysis.	03
2.	Explain the application of protein microarrays for the detection of post-translational protein phosphorylation.	03
3.	Explain the 2D-DIGE workflow for proteome separation and peptide mass fingerprinting for identification of differentially expressed protein spots.	03
4.	Explain the quantitative proteome expression profiling using isotope-coded affinity tags.	03
5.	Give an overview of multidimensional liquid chromatography-tandem mass spectrometry based separation and identification of proteins in proteome samples.	03
6.	Explain the identification of binary protein-protein interactions using an <i>In-vivo</i> library-based screening method.	03
7.	Explain the metabolic stable isotope coding strategy for quantitative proteome analysis.	03
8.	Describe the different categories of biological databases citing suitable examples.	03
9.	Describe various types of bioinformatics tools citing suitable examples.	03
10.	Differentiate between PAM and BLOSUM amino acid scoring matrices.	03
11.	Describe the terminologies used to characterize a phylogenetic tree.	03
12.	Expand the following terms: (i) PDB (ii) SCOP (iii) BLAST	03